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AXLE LOCK FOR COUPLING A ROLLER TO A FRAME

Abstract

An axle lock for coupling a roller to a frame is disclosed. The axle lock comprises a locking plate having a first locking plate aperture and a second locking plate aperture. Further, the first locking plate aperture allows an axle of the roller to be positioned within a cam handle. A bolt extending through the second locking plate aperture, the bolt having a first portion having a plurality of threads, a second portion having a pin aperture. Further, a pin is positioned within the cam aperture and the pin aperture, and a nut. Further, the locking plate is positioned between the nut and the cam handle. Further, the nut is configured to pass through a frame aperture of the frame. Further, pivoting the cam handle on the pin axis locks the locking plate to the frame when the frame is positioned between the nut and the locking plate.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] The present applications claims priority to and the benefit of U.S. Provisional Application No. 63/559,746, filed Feb. 29, 2024, the contents of which are hereby incorporated by reference in its entirety.

TECHNOLOGICAL FIELD

[0002] An example embodiment relates generally to locking mechanism, and more particularly, to an axle lock for coupling a conveyor roller to a frame.

BACKGROUND

[0003] Rollers or conveyor rollers are widely used components in various industrial applications. The rollers are employed for providing support and mobility to various conveyor systems and other moving elements. Typically, the rollers are constructed with a cylindrical shaped body that is encased around an axle. The axle of the rollers allows integration of the rollers with conveyor frames or side rails. The rollers are required to be locked with the conveyor frames via conventional axle locks. The conventional axle locks are bolted onto the conveyor frames. To install the conventional axle locks onto the conveyor frame, the installer has to hold a first portion of the axle lock that is positioned on a first side of the conveyor frame while simultaneously manipulating a second portion of the axle lock that is positioned on an opposite, second side of the conveyor frame, which may be difficult.

[0004] Applicant has identified numerous areas of improvement in the existing technologies and processes, which are the subjects of embodiments described herein. Through applied effort, ingenuity, and innovation, many of these deficiencies, challenges, and problems have been solved by developing solutions that are included in embodiments of the present disclosure, some examples of which are described in detail herein.

BRIEF SUMMARY OF THE INVENTION

[0005] The following presents a summary of some example embodiments to provide a basic understanding of some aspects of the present disclosure. This summary is not an extensive overview and is intended to neither identify key or critical elements nor delineate the scope of such elements. It will also be appreciated that the scope of the disclosure encompasses many potential embodiments in addition to those here summarized, some of which will be further described in the detailed description that is presented later.

[0006] In an exemplary embodiment, an axle lock for coupling a roller to a frame is disclosed. The axle lock comprises a locking plate having a first locking plate aperture and a second locking plate aperture. Further, the first locking plate aperture is configured to allow an axle of the roller to be positioned within the locking plate. Further, a cam handle having a cam aperture and a bolt extending through the second locking plate aperture, where the bolt defining a bolt axis. Further, the bolt defining a first portion having a plurality of threads, and a second portion having a pin aperture. Further, a pin defining a pin axis where the pin is positioned within the cam aperture of the cam handle and the pin aperture of the bolt. Further, a nut coupled to the first portion of the bolt. Further, the locking plate is positioned between the nut and the cam handle. Further, the nut is configured to pass through a frame aperture of the frame. Further, pivoting the cam handle on the pin axis locks the locking plate to the frame when the frame is positioned between the nut and the locking plate.

[0007] In some embodiments, the first locking plate aperture and the second locking plate aperture are fabricated over the locking plate. Further, the first locking plate aperture and the second locking plate aperture having a predefined radius. In some embodiments, the locking plate is a first locking plate and the axle lock further comprises a second locking plate that has the first locking plate aperture and the second locking plate aperture, respectively, each of the first locking plate and the

second locking plate has the first locking plate aperture and the second locking plate aperture that are positioned linearly over each other with a predefined distance between the first locking plate aperture and the second locking plate aperture. Further, the first locking plate and the second locking plate are parallelly coupled such that each of the first locking plate aperture and the second locking plate aperture on the first locking plate and the second locking plate superimpose.
[0008] In some embodiments, rotating the cam handle on the bolt axis locks the locking plate to the frame when the frame is positioned between the nut and the locking plate. Further, rotating the cam handle on the bolt axis facilitates the plurality of threads to rotate inside the nut that allows tightening of the nut with the bolt.
[0009] In some embodiments, a non-threaded shank is fabricated in between the plurality of threads at the first portion of the bolt. Further, a plurality of balls are positioned within the nut. Further, rotating the cam handle on the bolt axis pushes the plurality of balls in a radially outward direction relative to the nut, thereby locking the nut with the frame. Further, the nut corresponds to a hex nut, and the nut comprises a plurality of nut apertures that are each configured to allow a corresponding ball to be positioned at least partially within.
[0010] In some embodiments, at least one washer sandwiched between the cam handle and the locking plate, and configured to distribute load of the bolt over the locking plate. Further, the at least one washer is tapered at an angle such that pivoting the cam handle on the pin axis locks rotational direction movement of the cam handle.
[0011] In an exemplary embodiment, a method is disclosed. The method comprises steps of positioning a locking plate over an axle of a roller via a first locking plate aperture of the locking plate. The method further comprises steps of extending a bolt through a second locking plate aperture of the locking plate. Further, the bolt defining a bolt axis, and comprises a first portion having a plurality of threads and a second portion having a pin aperture. The method further comprises steps of positioning a pin within a cam aperture of a cam handle and the pin aperture of the bolt. Further, the pin defining a pin axis. The method further comprises coupling a nut to the first portion of the bolt. Further, the locking plate is positioned between the nut and the cam handle. Furthermore, the nut is configured to pass through a frame aperture of a frame. Further, pivoting the cam handle on the pin axis locks the locking plate to the frame when the frame is positioned between the nut and the locking plate.
[0012] The above summary is provided merely for purposes of summarizing some example embodiments to provide a basic understanding of some aspects of the present disclosure. Accordingly, it will be appreciated that the above-described embodiments are merely examples and should not be construed to narrow the scope or spirit of the present disclosure in any way. It will be appreciated that the scope of the present disclosure encompasses many potential embodiments in addition to those here summarized, some of which will be further described below.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0013] Having thus described certain example embodiments of the present disclosure in general terms, reference will hereinafter be made to the accompanying drawings, which are not necessarily drawn to scale, and wherein:
[0014] FIG. **1** illustrates an exploded view of an axle lock for coupling a roller to a frame in accordance with an example embodiment the present disclosure;
[0015] FIG. **2** illustrates a side view of the axle lock in accordance with the example embodiment of the present disclosure; and
[0016] FIGS. **3**A-**3**E illustrate a sequential process of installation of the axle lock over an axle of the roller for coupling the roller to the frame in accordance with an example embodiment of the

present disclosure.

DETAILED DESCRIPTION

[0017] Some embodiments will now be described more fully hereinafter with reference to the accompanying drawings, in which some, but not all, embodiments of the present disclosure are shown. Indeed, various embodiments may be embodied in many different forms and should not be construed as limited to the embodiments set forth herein; rather, these embodiments are provided so that this disclosure will satisfy applicable legal requirements.

[0018] The components illustrated in the figures represent components that may or may not be present in various embodiments of the present disclosure described herein such that embodiments may include fewer or more components than those shown in the figures while not departing from the scope of the present disclosure. Some components may be omitted from one or more figures or shown in dashed line for visibility of the underlying components.

[0019] As used herein, the term “comprising” means including but not limited to and should be interpreted in the manner it is typically used in the patent context. Use of broader terms such as comprises, includes, and having should be understood to provide support for narrower terms such as consisting of, consisting essentially of, and comprised substantially of.

[0020] The phrases “in various embodiments,” “in one embodiment,” “according to one embodiment,” “in some embodiments,” and the like generally mean that the particular feature, structure, or characteristic following the phrase may be included in at least one embodiment of the present disclosure and may be included in more than one embodiment of the present disclosure (importantly, such phrases do not necessarily refer to the same embodiment).

[0021] The word “example” or “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any implementation described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other implementations.

[0022] If the specification states a component or feature “may,” “can,” “could,” “should,” “would,” “preferably,” “possibly,” “typically,” “optionally,” “for example,” “often,” or “might” (or other such language) be included or have a characteristic, that a specific component or feature is not required to be included or to have the characteristic. Such a component or feature may be optionally included in some embodiments or it may be excluded.

[0023] Embodiments of the present disclosure will be described more fully hereinafter with reference to the accompanying drawings in which like numerals represent like elements throughout the several figures, and in which example embodiments are shown. Embodiments of the present disclosure may, however, be embodied in alternative forms and should not be construed as being limited to the embodiments set forth herein. The examples set forth herein are non-limiting examples and are merely examples among other possible examples.

[0024] The present disclosure discloses an axle lock for coupling a conveyor roller to a frame. The present disclosure provides various embodiments for locking an axle of a roller. Embodiments may be configured to install and remove the roller from the frame by rotating a cam handle. Embodiments may be configured to lock movement of the cam handle by using a bolt and a nut.

[0025] FIG. **1** illustrates an exploded view of an axle lock **100** for coupling a roller **102** to a frame **104**, in accordance with an example embodiment of the present disclosure. FIG. **2** illustrates a side view of the axle lock **100**, in accordance with the example of the present disclosure.

[0026] In some embodiments, the axle lock **100** may comprise a locking plate **106**. The locking plate **106** may comprise a first locking plate **108**A and a second locking plate **108**B, and a first locking plate aperture (not shown). The axle lock **100** may comprise a bolt **110**, a cam handle **112**, and a nut **114**. In some embodiments, the locking plate **106** may be configured to be mounted over an axle **116** of the roller **102** via the first locking plate aperture. The first locking plate aperture may be a hole fabricated through a thickness of the locking plate **106**, which may include the first locking plate **108**A and the second locking plate **108**B. The first locking plate aperture may allow the axle **116** of the roller **102** to pass through the locking plate **106**. Further, the roller **102** may be employed in various fields of applications. In an exemplary embodiment, the roller **102** may be integrated within a conveyor system to move one or more items or packages over the conveyor system. In some embodiments, the roller **102** may be installed with the frame **104** of the conveyor system.

[0027] In some embodiments, the roller **102** may comprise the axle **116** and an outer shell **118**. In some embodiments, the outer shell **118** may configured to be in direct contact with the one or more packages that are carried by the roller **102**. In some embodiments, the axle **116** of the roller **102** may be configured to allow coupling of the roller **102** with the frame **104** via the first locking plate aperture. Further, the axle **116** may be integrated with the frame **104** through a frame aperture (not shown) that extends through the frame **104**. In some embodiments, the locking plate **106** may be configured to be positioned parallel to the frame **104**. In some embodiments, the locking plate **106** may be crafted with a planar surface (not shown). Further, the planar surface of the locking plate **106** may allow proper positioning and alignment of the locking plate **106** with the frame **104**. In some embodiments, the locking plate **106** may be crafted with a material such as, but is not limited to, steel, aluminum, alloys, or any other material known in the art. In some embodiments, the locking plate **106** may define a length L, width W, and thickness T.

[0028] In some embodiments, the locking plate **106** may comprise the first locking plate **108**A and the second locking plate **108**B. In some embodiments, the first locking plate **108**A and the second locking plate **108**B may be crafted with the first locking plate aperture (not shown). Further, the first locking plate **108**A and the second locking plate **108**B may be crafted with a second locking plate aperture (not shown). In some embodiments, each of the first locking plate aperture and the second locking plate aperture may be positioned linearly over each other (e.g., aligned vertically) with a predefined distance. In some embodiments, the predefined distance between the first locking plate aperture and the second locking plate aperture may define a length L. In some embodiments, the first locking plate **108**A and the second locking plate **108**B may be parallelly coupled such that each of the first locking plate aperture on the first locking plate **108**A and the first locking plate aperture on the second locking plate **108**B superimpose (e.g., align horizontally).

[0029] In some embodiments, the first locking plate aperture may be configured to receive the axle **116** during installation of the roller **102** with the frame **104**. In some embodiments, the first locking plate aperture may be crafted with a shape that may be selected from a group of shapes such as, but are not limited to, circular shape, square shape, hexagonal shape, or any other shape known in the art. Further, the first locking plate aperture may have a predefined radius. In some embodiments, the locking plate **106** may be configured to enable a secured positioning of the frame **104** between the locking plate **106** and the roller **102**.

[0030] In some embodiments, the bolt **110** may be engaged with the locking plate **106** via at least one washer **120**. In some embodiments, the bolt **110** may comprise a first portion **122** and a second portion **124**. Further, the first portion **122** may be fabricated with a plurality of threads **126**. In some embodiments, the bolt **110** may define a bolt axis (not shown). In some embodiments, the plurality of threads **126** may be configured to allow fastening of the bolt **110** through the second locking plate aperture, upon rotation of the cam handle **112** on the bolt axis.

[0031] In some embodiments, the second portion **124** may be crafted with a pin aperture **128**. In some embodiments, the pin aperture **128** may be configured to receive a pin **130** that may be engaged within the pin aperture **128**. Further, the pin **130** may define a pin axis (not shown). The pin axis may extend orthogonally from the bolt axis. Further, the pin aperture **128** may be crafted with a shape that may comprise at least one of square shape, circular shape, rectangular shape, or any other shape known in the art. In some embodiments, the pin **130** may be configured to enable coupling of the cam handle **112** with the pin aperture **128** of the bolt **110**. In some embodiments, the cam handle **112** may comprise a cam aperture **132**. Further, the cam aperture **132** may be configured to enable coupling of the cam handle **112** with the pin **130**. In some embodiments, the cam handle **112** may be pivoted by a user around the pin axis.

[0032] In some embodiments, the locking plate **106** may be monolithic with at least one washer **120**. The at least one washer **120** may be sandwiched between the cam handle **112** and the locking plate **106**. In some embodiments, the at least one washer **120** may be configured to distribute load of the bolt **110** over the locking plate **106** to allow independent rotational movement of the bolt **110**. Further, the at least one washer **120** may be tapered at an angle, as illustrated by **134**. In some embodiments, the at least one washer **120** may be configured to interlock the cam handle **112** when the cam handle **112** is pivoted over the pin axis to prevent further movement of the cam handle **112**.
[0033] In some embodiments, the axle lock **100** may further comprise the nut **114**. In some embodiments, the nut **114** may be positioned parallel to the second locking plate aperture machined on the locking plate (e.g., the second locking plate apertures of the first locking plate **108**A and the second locking plate **108**B). In some embodiments, the nut **114** may be coupled with the first portion **122** of the bolt **110** via the plurality of threads **126**. In some embodiments, the nut **114** may be fastened with the first portion **122** of the bolt **110** via the second locking plate aperture.
[0034] In some embodiments, the second locking plate aperture may be crafted with a shape that may be selected from a group of shapes such as, but are not limited to, circular shape, square shape, hexagonal shape, or any other shape known in the art. Further, the second locking plate aperture may have a predefined radius. In some embodiments, the second locking plate aperture may be crafted on the first locking plate **108**A and the second locking plate **108**B of the locking plate **106**. Further, the first locking plate **108**A and the second locking plate **108**B may be parallelly placed such that the second locking plate aperture may superimpose with the frame aperture crafted on the frame **104** (e.g., the second locking plate aperture may be aligned horizontally with the frame aperture). In some embodiments, the rotations of the cam handle **112** over the pin axis facilitates in tightening of the bolt **110** with the nut **114**.
[0035] In some embodiments, the nut **114** may correspond to a hex nut. Further, the nut **114** may comprise a plurality of nut apertures **136**. In some embodiments, the plurality of nut apertures **136** of the nut **114** may be machined to accommodate a plurality of balls **138**. In some embodiments, the plurality of nut apertures **136** may be configured to allow a corresponding ball **138** to be positioned within the nut **114**. In some embodiments, the bolt **110** may be configured to push the plurality of balls **138** in a radially outwards direction during the rotation of the cam handle **112** on the bolt axis. In some embodiments, the bolt **110** may comprise a non-threaded shank **140**. Further, the non-threaded shank **140** may be configured to push the plurality of balls **138** in the radially outwards direction from the nut **114**, thereby locking the nut **114** with the frame **104**.
[0036] FIGS. **3**A-**3**E illustrate a sequential process of installation of the axle lock **100** over the axle **116** of the roller **102** for coupling the roller **102** to the frame **104**, in accordance with the example embodiment of the present disclosure. FIGS. **3**A-**3**E are described in conjunction with FIGS. **1-2**.
[0037] In some embodiments, the locking plate **106** may be configured to mount over the axle **116** of the roller **102**. In some embodiments, the axle **116** of the roller **102** may rest on the frame aperture fabricated with the frame **104**. In some embodiments, the first locking plate **108**A and the second locking plate **108**B of the locking plate **106** may be crafted with the first locking plate aperture. In some embodiments, the first locking plate aperture may allow coupling of the locking plate **106** with the axle **116** of the roller **102** as illustrated in FIG. **3**A. Further, the first locking plate **108**A and the second locking plate **108**B of the locking plate **106** may be crafted with the second locking plate aperture vertically placed below the first locking plate aperture. The second locking plate aperture may be configured to allow the nut **114** pass through the locking plate **106** as illustrated in FIG. **3**B.
[0038] In some embodiments, the bolt **110** may be attached with the locking plate **106** via the at least one washer **120**. In some embodiments, the bolt **110** may comprise the first portion **122** and the second portion **124**. In some embodiments, the first portion **122** may be crafted with the plurality of threads **126**. In some embodiments, the bolt **110** may be configured to extend through the second locking plate aperture and couple to the nut **114**. Further, the bolt **110** may be configured

to define the bolt axis.
[0039] In some embodiments, the second portion **124** of the bolt **110** may be constructed with the pin aperture **128**. Further, the pin aperture **128** may be configured to accommodate the pin **130** through a machined process. In some embodiments, the pin **130** may define the pin axis. In some embodiments, the pin **130** may be configured to couple the cam handle **112** with the second portion **124** of the bolt **110**.
[0040] In some embodiments, the at least one washer **120** may be sandwiched between the bolt **110** and the locking plate **106**. In some embodiments, the at least one washer **120** may be configured to distribute load of the bolt **110** over the locking plate **106**. In some embodiments, the at least one washer **120** may be crafted with a hole. In some embodiments, the radius of the at least one washer **120** may be selected in a view of enable proper movement of the bolt **110** through the at least one washer **120**.
[0041] Further, as illustrated in FIG. **3**C, the cam handle **112** may be configured to rotate in either clockwise rotational direction or in anticlockwise rotational direction around the bolt axis, as illustrated by an arrow **302**. In some embodiments, the cam handle **112** may be configured to enable the user to provide movement to the cam handle **112** in either the clockwise rotational direction or in the anticlockwise rotational direction around the bolt axis, which may allow fastening of the bolt **110** through the second locking plate aperture. Further, the cam handle **112** may be crafted with a shape that may be selected from a group of shapes such as, but are not limited to, an elliptical shape, parabolic shape, sinusoidal shape, or any other shape known in the art.
[0042] In some embodiments, the nut **114** may be coupled with the first portion **122** of the bolt **110**. In some embodiments, the first portion **122** of the bolt **110** may be crafted with the plurality of threads **126**. Further, the fastening of the bolt **110** through the second locking plate aperture may facilitate the plurality of threads **126** to fasten inside the nut **114**. In some embodiments, the fastening of the plurality of threads **126** inside the nut **114** may result in tightening of the nut **114** with the bolt **110** as illustrated in FIG. **3**C.
[0043] Further, the bolt **110** may comprise the non-threaded shank **140** as illustrated in FIGS. **1** and **2**. The non-threaded shank **140** may be positioned between the plurality of threads **126** and the pin aperture **128**. In some embodiments, the rotational movement of the bolt **110** facilitates a sequential movement of the non-threaded shank **140** inside the nut **114**. In some embodiments, the nut **114** may comprise the plurality of nut apertures **136**.
[0044] In some embodiments, the plurality of balls **138** may be positioned within the plurality of nut apertures **136**. Further, the movement of the cam handle **112** over the pin axis facilitates movement of the non-threaded shank **140** inside the nut **114**. Further, the non-threaded shank **140** may be configured to push the plurality of balls **138** in the radially outwards direction from the nut **114** as illustrated in FIG. **3**D. In some embodiments, the movement of the plurality of balls **138** in the radially outwards direction relative to the nut **114** may facilitate locking of the nut **114** with the frame **104**. For example, the movement of the plurality of balls **138** in the radially outwards direction may prevent the nut **114** from being able to pass through the frame aperture on the frame **104**.
[0045] In some embodiments, the first portion **122** of the bolt **110** may be crafted with a circular groove. In some embodiments, the circular groove may be configured to enable snap-fitting of the plurality of balls **138** after successful fastening of the bolt **110** inside the nut **114**. Further, the circular groove may be crafted with one or more elliptical boundaries to enable snap-fitting of the plurality of balls **138** inside the circular groove thereby locking the bolt **110** inside the nut **114**.
[0046] In some embodiments, the at least one washer **120** may be tapered at the angle, as illustrated by **134**. In some embodiments, the cam handle **112** may be pivoted on the pin axis, as illustrated by an arrow **304**, to lock the locking plate **106** to the frame **104**. In some embodiments, the at least one washer **120** may be configured to interlock with the cam handle **112** upon pivoting of the cam

handle **112**. In some embodiments, the pivoting of the cam handle **112** facilitates locking of rotational direction of the cam handle **112** as illustrated in FIG. **3**E. Further, the pivoting of the cam handle **112** facilitates in locking of the locking plate **106** with the frame **104** when the frame **104** is positioned between the nut **114** and the locking plate **106**.

[0047] In an example embodiment, a method is disclosed. The method comprises one or more operations that may be facilitate locking of the axle **116** of the roller **102** with the frame **104**. In some embodiments, the locking plate **106** may be positioned over the axle **116** of the roller **102** by the first locking plate aperture of the locking plate **106**. For example, the locking plate **106** is engaged with the axle **116** of an idler roller that is further integrated within a conveyor system. Further, the locking plate **106** is machined with the first locking plate aperture that facilitates engaging of the locking plate **106** with the axle **116** of the idler roller.

[0048] In some embodiments, the bolt **110** may be extended through the second locking aperture of the locking plate **106**. Further, the bolt **110** may be configured to define the bolt axis. In some embodiments, the bolt **110** may comprise the first portion **122** and the second portion **124**. Further, the first portion **122** may comprise the plurality of threads **126** and the second portion **124** may comprise the pin aperture **128**. For example, the locking plate **104** is machined with a second locking plate aperture that allows extending of a bolt **110** through the second locking plate aperture of the locking plate **108**. Further the bolt **110** comprises a first portion **122** and a second portion **124**. The first portion **122** of the bolt **110** is crafted with a plurality of threads **126** and the second portion **124** of the bolt **110** is crafted with a pin aperture **128**.

[0049] In some embodiments, the pin **130** may be positioned within the cam aperture **132** of the cam handle **112** and the pin aperture **128** of the bolt **110**. Further, the pin **130** may define the pin axis. For example, the pin aperture **128** of the bolt **110** is positioned with the pin **130** upon passing through the cam aperture **132** of a cam handle **112** to secure the cam handle **112** with the bolt **110**. Further, the pin **130** defines a pin axis.

[0050] In some embodiments, the nut **114** may be coupled with the first portion **122** of the bolt **110**. Further, the locking plate **106** may be positioned between the nut **114** and the cam handle **112**. Furthermore, the nut **114** may be configured to pass through the frame aperture of the frame **104**. Further, the cam handle **112** may be pivoted on the pin axis to lock the locking plate **106** to the frame **104** when the frame **104** may be positioned between the nut **114** and the locking plate **106**.

[0051] Embodiments may be configured to provide a secured locking of the axle **116** of the roller **102** with the frame **104**. Embodiments may ensure alignment of the roller **102** with the frame **104** by providing the secured locking of the axle **116** with the frame **104**. Embodiments may ensure proper fitting of the axle **116** with the frame **104** by means of the bolt **110** and the nut **114**.

[0052] Embodiments may ensure proper locking of the axle **116** by interlocking the cam handle **112** with the at least one washer **120**. Embodiments may allow easy installation of the roller **102** with the frame **104**. Embodiments may avoid any manual involvement during installation of the roller **102**. Embodiments may ensure proper alignment and fitting of the roller **102** with the frame **104**. Embodiments may provide replacement for conventional bolted lock mechanism. Embodiments may enhance durability of the roller **102**.

[0053] Embodiments may allow installation of the axle lock **100** with access to only one side of the frame **104** of the conveyor. For example, and as discussed, the axle lock **100** may be installed on the frame **104** by passing the nut **114** through the frame aperture. Once the nut **114** is on the opposite side of the frame **104**, the cam handle **112** may be rotated on the bolt axis and/or pivoted on the pin axis, which may move the balls **138** radially outward, which may prevent the nut **114** from passing through the frame aperture. Preventing the nut **114** from passing through the frame aperture, along with tightening the cam handle **112**, may lock the axle lock **100** onto the frame **104**. Notably, conventional axle locks often require access to both sides of the frame of the conveyor by requiring an installer to hold a portion of the axle lock that is positioned on a side of the frame while simultaneously tightening a nut that is on an opposite side of the frame, which may be difficult. As such, the axle lock **100** of the present disclosure may be easier to install than conventional axle locks.

[0054] Many modifications and other embodiments of the inventions set forth herein will come to mind to one skilled in the art to which these inventions pertain having the benefit of the teachings presented in the foregoing descriptions and the associated drawings. Therefore, it is to be understood that the inventions are not to be limited to the specific embodiments disclosed and that modifications and other embodiments are intended to be included within the scope of the appended claims. Moreover, although the foregoing descriptions and the associated drawings describe example embodiments in the context of certain example combinations of elements and/or functions, it should be appreciated that different combinations of elements and/or functions may be provided by alternative embodiments without departing from the scope of the appended claims. In this regard, for example, different combinations of elements and/or functions than those explicitly described above are also contemplated as may be set forth in some of the appended claims. Although specific terms are employed herein, they are used in a generic and descriptive sense only and not for purposes of limitation.

Claims

1. An axle lock for coupling a roller to a frame, the axle lock comprising: a locking plate having a first locking plate aperture and a second locking plate aperture, wherein the first locking plate aperture is configured to allow an axle of the roller to be positioned within the locking plate; a cam handle having a cam aperture; a bolt extending through the second locking plate aperture, the bolt defining a bolt axis, the bolt having: a first portion having a plurality of threads; and a second portion having a pin aperture; a pin defining a pin axis, wherein the pin is positioned within the cam aperture of the cam handle and the pin aperture of the bolt; and a nut coupled to the first portion of the bolt, wherein the locking plate is positioned between the nut and the cam handle, wherein the nut is configured to pass through a frame aperture of the frame, wherein pivoting the cam handle on the pin axis locks the locking plate to the frame when the frame is positioned between the nut and the locking plate.

2. The axle lock of claim 1, wherein the first locking plate aperture and the second locking plate aperture are fabricated over the locking plate, wherein the first locking plate aperture and the second locking plate aperture having a predefined radius.

3. The axle lock of claim 1, wherein the locking plate is a first locking plate and the axle lock further comprises a second locking plate that has a first locking plate aperture and the second locking plate aperture, respectively, each of the first locking plate and the second locking plate has the first locking plate aperture and the second locking plate aperture that are positioned linearly over each other with a predefined distance between the first locking plate aperture and the second locking plate aperture.

4. The axle lock of claim 3, wherein the first locking plate and the second locking plate are parallelly coupled such that each of the first locking plate aperture and the second locking plate aperture on the first locking plate and the second locking plate superimpose.

5. The axle lock of claim 1, wherein rotating the cam handle on the bolt axis locks the locking plate to the frame when the frame is positioned between the nut and the locking plate.

6. The axle lock of claim 5, wherein rotating the cam handle on the bolt axis facilitates the plurality of threads to rotate inside the nut that allows tightening of the nut with the bolt.

7. The axle lock of claim 1, wherein a non-threaded shank is fabricated in between the plurality of threads at the first portion of the bolt.

8. The axle lock of claim 7, wherein a plurality of balls are positioned within the nut, wherein rotating the cam handle on the bolt axis pushes the plurality of balls in a radially outward direction relative to the nut, thereby locking the nut with the frame.

9. The axle lock of claim 8, wherein the nut corresponds to a hex nut, and the nut comprises a plurality of nut apertures that are each configured to allow a corresponding ball to be positioned at least partially within.
10. The axle lock of claim 5, further comprising at least one washer sandwiched between the cam handle and the locking plate, and configured to distribute load of the bolt over the locking plate, and wherein the at least one washer is tapered at an angle such that pivoting the cam handle on the pin axis locks rotational direction movement of the cam handle.
11. A method comprising: positioning a locking plate over an axle of a roller via a first locking plate aperture of the locking plate; extending a bolt through a second locking plate aperture of the locking plate, wherein the bolt defining a bolt axis, and comprises a first portion having a plurality of threads and a second portion having a pin aperture; and positioning a pin within a cam aperture of a cam handle and the pin aperture of the bolt, wherein the pin defining a pin axis; and coupling a nut to the first portion of the bolt, wherein the locking plate is positioned between the nut and the cam handle, wherein the nut is configured to pass through a frame aperture of a frame, wherein pivoting the cam handle on the pin axis locks the locking plate to the frame when the frame is positioned between the nut and the locking plate.
12. The method of claim 11, wherein the first locking plate aperture and the second locking plate aperture are fabricated over the locking plate, wherein the first locking plate aperture and the second locking plate aperture having a predefined radius.
13. The method of claim 11, wherein the locking plate is a first locking plate and the axle lock further comprises a second locking plate that has the first locking plate aperture and the second locking plate aperture, respectively, each of the first locking plate and the second locking plate has the first locking plate aperture and the second locking plate aperture that are positioned linearly over each other with a predefined distance between the first locking plate aperture and the second locking plate aperture.
14. The method of claim 13, wherein the first locking plate and the second locking plate are parallelly coupled such that each of the first locking plate aperture and the second locking plate aperture on the first locking plate and the second locking plate superimpose.
15. The method of claim 11, wherein rotating the cam handle on a bolt axis locks the locking plate to the frame when the frame is positioned between the nut and the locking plate, where rotating the cam handle on the bolt axis facilitates the plurality of threads to rotate inside the nut that allows tightening of the nut with the bolt.
16. The method of claim 15, wherein a non-threaded shank is fabricated in between the plurality of threads at the first portion of the bolt.
17. The method of claim 16, wherein a plurality of balls are positioned within the nut, wherein rotating the cam handle on the bolt axis pushes the plurality of balls in a radially outward direction relative to the nut, thereby locking the nut with the frame.
18. The method of claim 17, the nut corresponds to a hex nut, and the nut comprises a plurality of nut apertures that are each configured to allow a corresponding ball to be positioned at least partially within.
19. The method of claim 11, further comprising at least one washer sandwiched between the cam handle and the locking plate, and configured to distribute load of the bolt over the locking plate.
20. The method of claim 19, wherein the at least one washer is tapered at an angle such that pivoting the cam handle on the pin axis locks rotational direction movement of the cam handle.